Insights Log:

Visualization 1 - Rides by Hour:

Peak demand is at 5PM (17:00) - evening rush hour
Low demand between 1AM - 5AM - late night/early morning
Demand starts rising from 6AM onwards as people go to work

Visualization 2 – Rides by Day of Week:

Thursday has the highest Uber rides in NYC and Sunday has the lowest
Weekdays generally have more rides than weekends
This suggests Uber is used more for work commutes than leisure

Visualization 3 – Rides by Month:

September has the highest Uber rides
Clear upward trend from April to September – Uber was growing rapidly in NYC in 2014
Shows month over month growth in Uber adoption

Visualization 4 – Heatmap Insights:

Peak Demand (Darkest Red cells):

Tuesday, Wednesday, Thursday at 5PM-6PM (17:00) –

absolute peak demand
Friday evening (17:00-21:00) — high demand extends longer into night
Thursday 17:00 — darkest cell overall — highest single demand point

Weekend Pattern (Saturday & Sunday):

Saturday 0:00-1:00 (midnight) — surprisingly high — people returning from Friday night out
Sunday 0:00 — also high for same reason
Weekends have lower daytime demand but higher late night demand compared to weekdays

Low Demand (Light Yellow):

3AM-5AM across all days — minimum demand
Sunday morning — quietest time of the week

Key Business Insight:

Weekdays = commute driven demand — peaks at office hours
Weekends = nightlife driven demand — peaks after midnight

Visualization 5 — Rides by Base:

B02617 is the busiest dispatch center — handles most rides in NYC

5 total bases operating in NYC
Shows unequal distribution — some bases handle significantly more rides than others
Business insight: Uber should allocate more drivers to B02617 zone for optimal operations

Visualization 6 — Rush Hour Analysis:

33.1% of all rides happen during rush hours (7-9 AM and 5-7 PM)
66.9% rides happen outside rush hours
Business insight: Despite rush hour being only 6 hours out of 24, it contributes to 1 in 3 rides — showing extremely high demand concentration during these hours
Uber should implement surge pricing strategy during these windows

Visualization 7 — Monthly Growth Trend:

Consistent upward growth from April to September 2014
Rides grew from ~580,000 in April to ~1,000,000+ in September
That's nearly 80% growth in just 6 months — explosive adoption
Business insight: Uber was in rapid growth phase in NYC in 2014 — month over month demand kept increasing showing strong product market fit

Visualization 8 — Correlation Heatmap:

hour vs day = -0.00 — no correlation — hour of day and day of month are completely independent
hour vs month = -0.02 — no correlation — hour patterns don't change across months
day vs month = -0.01 — no correlation
Business insight: Ride demand by hour is consistent across all days and months — meaning peak hours (5PM) are always peak regardless of which day or month it is. This makes demand highly predictable for forecasting.

ML Model Evaluation:

MAE: 96.59 — model's predictions are off by only ~97 rides per hour on average
R^2 Score: 0.9423 — model explains 94.23% of variance in ride demand
Model Accuracy: 94.23% — very strong for a demand forecasting model

Visualization 9 — Actual vs Predicted:

Blue line = Actual rides
Orange dashed line = Predicted rides
Both lines follow almost identical patterns throughout

Model accurately captures peaks and valleys of demand
Minor deviations at extreme peaks but overall very tight fit
Confirms 94.23% accuracy visually — predictions are very close to reality

Visualization 10 — Feature Importance:

Hour is by far the most important feature (~0.6 importance score) — time of day drives Uber demand more than anything else
Weekday is second most important (~0.13) — which day of week matters
Month is third (~0.12) — season/month has some influence
Day of month is least important (~0.08) — specific date doesn't matter much
Business insight: To predict Uber demand, hour of day is the #1 factor — more important than day or month combined